Yuan-Kai (Kai) Liu

Postdoctoral Scholar
Doerr School of Sustainability | Geophysics
Stanford University
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Education

2019 – 2026 PhD, Geophysics (minor in Computational Science and Engineering)
California Institute of Technology (Caltech)
Thesis: Space Geodetic Constraints on Plate Motion, Fault Creep, and Megathrust Coupling
Advised by Mark Simons, Zachary Ross

2016 – 2018 MSc, Earth Science & Engineering
King Abdullah University of Science and Technology (KAUST), Saudi Arabia
Thesis: Ground deformation related to caldera collapse and ring-fault activity
Advised by Sigurjón Jónsson

2010 – 2015 BSc, Geological Sciences
BEng, Civil Engineering, National Taiwan University, Taiwan

Research Interests

Radar and GNSS spaceborne geodesy, Seismic cycle, Lithospheric deformation, Inverse problem, Seismicity

Appointments

2026.1 – present	Postdoctoral Scholar, Stanford University + Live Analysis of Volcanic Activity with SAR (NASA MEaSUREs project) + Spaceborne measures of soil moisture and groundwater deformation
2019 – 2025	Graduate Research, Caltech Seismological Laboratory, Caltech
2016 – 2018	Graduate Research, Crustal Deformation and InSAR Group, KAUST (Saudi Arabia)
2014 – 2015	Research Assistant, Institute of Earth Sciences, Academia Sinica (Taiwan)
2014	Visiting Student, Department of Earth Sciences, Tohoku University
2013	Project Intern, Carbon Capture and Storage Project, Taiwan Power Company
2012	Project Intern, Institute of Earth Sciences, Academia Sinica

Fellowships and Awards

2022	J. Yang Scholarship, Caltech, $15,000
2019 – 2020	Caltech C Scholarship, Caltech, $5,000
2019 – 2020	Division of Geological and Planetary Sciences Fellowship, Caltech
2017	Outstanding Student Poster Award (OSPP), European Geosciences Union
2016 – 2018	Graduate Program Fellowship in Earth Science and Engineering, KAUST
2014	National Taiwan University Dean's Award
2012	National Taiwan University Undergraduate Scholarship
2012	National Taiwan University Presidential Award

Teaching and Mentorship

2023	Mentor of Caltech Seismolab Earthquake Fellows: High school summer program
2022	Mentor of Caltech WAVE Fellows: 1 undergraduate summer research
2021	Mentor of Caltech SURF Fellows: 1 undergraduate summer research
2021 – 2022	TA at Caltech: Ge 271 – Dynamics of Seismicity (Zachary Ross) Ge 167 – Tectonic Geodesy (Mark Simons) Ge 193 – Imaging Radar and Applications (Howard Zebker)
2014 – present	Science communication in geophysics, Archilife Research Foundation, Taiwan
2012 – 2014	Middle School Science and Math tutor

Services

2022 – present	Journal Reviewer: Nature Communications Earth & Environment (Nature) Geophysical Research Letters (AGU) Journal of Geophysical Research Solid Earth (AGU) Earth and Planetary Science Letters (Elsevier)
2021 – 2022	Organizing Committee, Caltech Seismological Laboratory Seminar
2021	Student Poster Judge: Southern California Earthquake Center Meeting

Publications and Presentations

[*] Corresponding author, [†] Contributed equally

[10] **Liu, Y.-K.** [*], Mallick, R., & Simons, M. (in prep). From Lima to Santiago, inference of interseismic loading along the Peru-Chile megathrust.

[9] **Liu, Y.-K.** [*], Yunjun, Z., & Simons, M. (in prep). Constraining the extension and interseismic coupling along the southern Dead Sea Transform.

[8] Liang, C. [*], Li, X., Wang Y., Liu, Z., Simons, M., Fielding, E. J., **Liu, Y.-K.**, Yun, S.-H. (in review). Evaluating ALOS-2 L-Band InSAR Observations of Tectonic Motion in the Western United States. Earth and Space Science (AGU).

[7] Pereiaslov, L. [*], Ragon, T., **Liu, Y.-K.**, Bato, M. G., Govorcin, M., Fielding, E., & Simons, M. (accepted on Sep 28, 2026). Bayesian co- and post-seismic slip model of the 2023 Mw 7.8 and Mw 7.7 Türkiye-Syria earthquake doublet from geodetic data. *Seismological Research Letters*.

[6] Wilding, J. [*], Wang, T., **Liu, Y.-K.**, & Ross, Z. E. (2026). Cascading magmatic unrest and aseismic faulting in the Aegean Sea enabled by elastic stress relay. *Science Advances, 12(39)*, eaec3198. https://doi.org/10.1126/sciadv.aec3198

[5] **Liu, Y. -K.**[*], Yunjun, Z., & Simons, M. (2025). Inferring tectonic plate rotations from InSAR time series. *Geophysical Research Letters, 52(12)*, e2025GL115137. https://doi.org/10.1029/2025GL115137

[4] Stephenson, O. L.[†], **Liu, Y.-K.**[†*], Yunjun, Z., Simons, M., Rosen, P., & Xu, X. (2022). The impact of plate motions on long-wavelength InSAR-derived velocity fields. *Geophysical Research Letters, 49*, e2022GL099835. https://doi.org/10.1029/2022GL099835

[3] **Liu, Y.-K.***, Ross, Z. E., Cochran, E. S., & Lapusta, N. (2022). A unified perspective of seismicity and fault coupling along the San Andreas Fault. *Science Advances, 8,* eabk1167. *https://doi.org/10.1126/sciadv.abk1167*

[2] Aldaajani, T., Simons, M., Yunjun, Z., Bekaert, D., Almalki, K.A., **Liu, Y.-K**. Using InSAR Time Series to Monitor Surface Fractures and Fissures in the Al-Yutamah Valley, Western Arabia. *Remote Sens.* 2022, *14*, 1769. https://doi.org/10.3390/rs14081769

[1] **Liu, Y.-K.**, Ruch, J., Vasyura-Bathke, H., & Jónsson, S. (2019). Influence of ring faulting in localizing surface deformation at subsiding calderas. *Earth and Planetary Science Letters, 526,* 115784. https://doi.org/10.1016/j.epsl.2019.115784

Invited Talks:

American Geophysical Union, Fall Meeting, New Orleans, 2025.
Title: Inferring Tectonic Plate Rotations from InSAR time series

Active Tectonics and Earthquake Hazards,
Lecture guest seminar, Middlebury College in Vermont, 2024
Title: A unified perspective of seismicity and fault coupling along the San Andreas Fault

American Geophysical Union, Fall Meeting, Chicago, 2022.
Title: The Impact of Plate Motions on Long-wavelength InSAR-derived velocity fields

Southern California Earthquake Hazards Assessment,
USGS Pasadena, Weekly Meeting, 2022.
Title: A unified perspective of seismicity and fault coupling along the San Andreas Fault

Conference Abstracts:

Decadal Interseismic Loading of the Peru-Chile Subduction Zone using wide-area InSAR coverage and Bayesian Inference
Liu, Y.-K., Pereiaslov, L., Mallick, R., Simons, M.
SZ4D Science Community Meeting, 2026 April, Long Beach, CA.

Inferring Tectonic Plate Rotations from InSAR time series
Liu, Y.-K., Yunjun, Z., Simons, M.
American Geophysical Union (AGU), 2025 Fall Meeting, New Orleans, CA.

High-Resolution Interseismic Deformation of the Southern Dead Sea Transform Fault
Liu, Y.-K., Yunjun, Z., Zheng, Y., Liang, C., Alotaibi, T., Simons, M.
American Geophysical Union (AGU), 2023 Fall Meeting, San Francisco, CA.

Imaging Deformation Processes along the Southern Dead Sea Transform using 8 years of InSAR
Liu, Y.-K., Stephenson, O. L., Yunjun, Z., Simons, M.
GAGE/SAGE 2023 Community Science Workshop, Pasadena, CA.

The Impact of Plate Motions on Long-Wavelength InSAR-Derived Velocity Fields
Liu, Y.-K., Stephenson, O. L., Yunjun, Z., Simons, M., Rosen, P., & Xu, X.
American Geophysical Union (AGU), 2022 Fall Meeting, Chicago, IL.

High-Resolution Interseismic Deformation of the Southern Dead Sea Transform Fault
Liu, Y.-K., Stephenson, O. L., Yunjun, Z., Simons, M.

American Geophysical Union (AGU), 2022 Fall Meeting, Chicago, IL.

A unified perspective of seismicity and fault coupling along the San Andreas Fault
Liu, Y.-K., Ross, Z. E., Cochran, E. S., Lapusta, N.
American Geophysical Union (AGU), 2021 Fall Meeting, New Orleans, LA.

A unified perspective of seismicity and fault coupling along the San Andreas Fault
Liu, Y.-K., Ross, Z. E., Cochran, E. S., Lapusta, N.
Poster #089, SCEC Contribution #11496. 2021 SCEC Annual Meeting, Palm Springs, CA.

The problem of ghost magma chambers under calderas
Liu, Y.-K., Ruch, J., Vasyura-Bathke, H., Jonsson, S.
Cities on Volcanoes 10 Meeting (2018), Naples, Italy.

Ring-fault activity at subsiding calderas studied from analogue experiments and numerical modeling
Liu, Y.-K., Ruch, J., Vasyura-Bathke, H., Jonsson, S.
American Geophysical Union (AGU), 2017 Fall Meeting, New Orleans, LA.

Contemporaneous ring fault activity and surface deformation at subsiding calderas studied using analogue experiments
Liu, Y.-K., Ruch, J., Vasyura-Bathke, H., Jonsson, S.
European Geosciences Union (EGU), General Assembly 2017, Vienna, Austria.

Numerical simulation study of Changhua and Chelungpu faults earthquake scenarios,
Lin, C.-H., T.-C. Lin, Y.-R. Chang, **Y.-K. Liu,** S.-J. Lee, Twin of Chi-Chi earthquake?
2012 Undergraduate summer project, Academia Sinica, Taiwan.

Software & Libraries

[4] Python Plate Motion Model (PyPMM). Liu et al.
[3] Miami InSAR Time-series software in Python (MintPy). Yunjun et. al., 2019
[2] Interferometric synthetic aperture radar Scientific Computing Environment (ISCE). Rosen et. al., 2012
[1] Python-based Atmospheric Phase Screen estimation (PyAPS). Jolivet et al., 2011